

Integrating GitLab with Moodle

Bachelor Thesis — HEIG-VD

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Improving developer workflows in computer science education

The Problem

Moodle and GitLab are both heavily used in computer science education, but operate independently. This separation creates fragmented workflows for teachers and students.

Teachers

- Create repositories manually
- Configure GitLab permissions by hand
- Manage multiple platforms

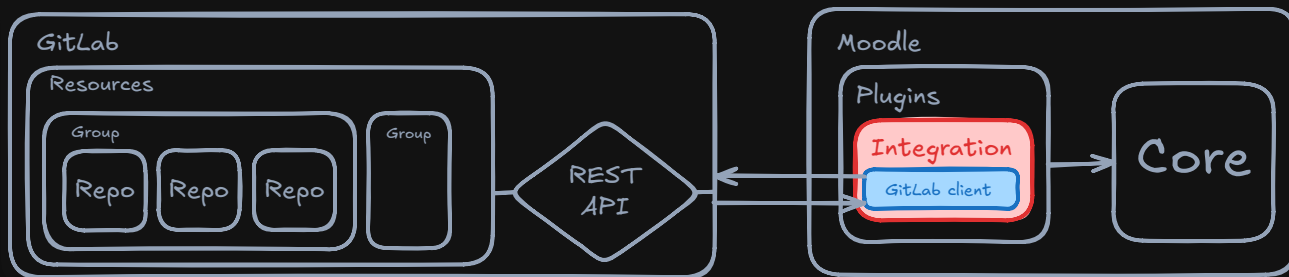
Students

- Inconsistent workflows between courses
- Switch between LMS and VCS
- Grades, deadlines, repositories in different places

Current solutions are either proprietary, GitHub-focused, or poorly integrated with Moodle.

Proposed Solution

A native Moodle plugin that integrates directly with GitLab through a dedicated PHP API client



Automation

Create and manage GitLab resources from Moodle

Unified Experience

Centralized assignments, repositories, grades, and feedback

Extensible Design

Modular architecture allowing future support for other VCS providers

State of the Art

No open-source solution currently provides a complete Moodle-native integration with GitLab

Solution	Open Source	Moodle Native	GitLab Support
GitHub Classroom	✗	✗	✗
Zohoflow	✗	✗	✓
Classmoji	✓	✗	✗
Canvas Gitlab Integration	✓	✗	✓
This Project	✓	✓	✓

The project aims to fill this gap with a native fully integrated and extensible open-source solution

Proof of Concept

A Moodle module plugin to demonstrate the integration idea.

Features

- Add a GitLab activity module
- Set a GitLab API token per course
- Automatically create groups / repositories for students
- Access repositories from Moodle
- Display created repositories list

Infrastructure as Code

- Terraform for infrastructure provisioning
- Ansible for automated configuration
- Authentik for authentication
- GitHub Actions for continuous deployment

Project Progress

Completed

- Project specifications and planning
- State-of-the-art research
- Moodle proof-of-concept module plugin
- Cloud infrastructure
- GitLab REST API PHP client
- Intermediate thesis report

Next Steps

- Teacher dashboard
- Student group selection / overview
- Advanced GitLab automation (webhooks, merge requests)

The technical foundation is complete.
The project is now entering its main implementation phase.

Demonstration